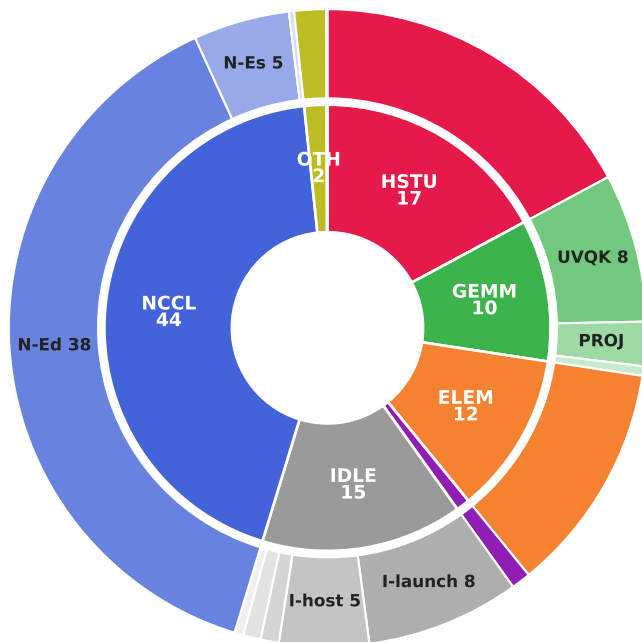
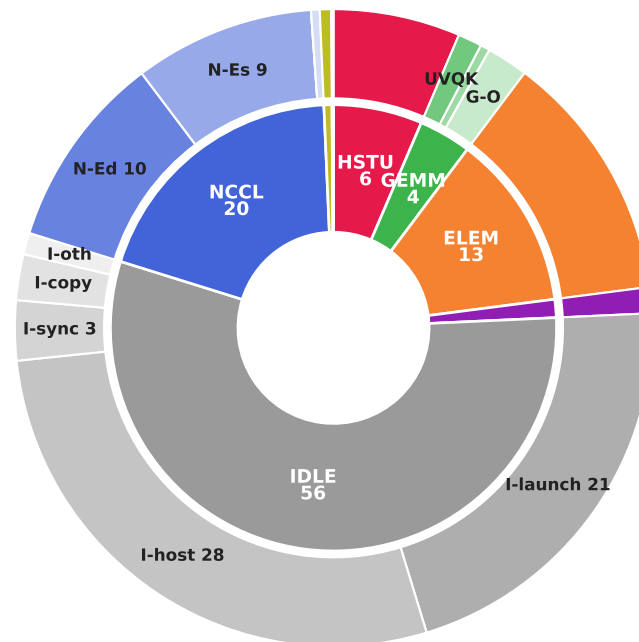


HSTU GPU-time sunburst — jagged runs, exposed accounting, fastest step (upstream PERF_ANALYSIS §2.2 method) inner ring = grouping · outer ring = leaves · % of the step

Our H100 · step 153 · 93.5 ms · D256 (jagged)



Our GB300 · step 155 · 65.5 ms · D128 (jagged)



- HSTU = hstu fwd/bwd
- UVQK = gemm / uvqk
- PROJ = gemm / projection
- G-O = gemm / others
- ELEM = elementwise
- EMB = embedding op
- N-Ed = nccl DENSE/all-reduce (exposed)
- N-Es = nccl SPARSE/all-to-all (exposed)
- N-Os = nccl sparse (overlap)
- I-launch = idle: kernel-launch (CPU dispatch)
- I-host = idle: host/python gap
- I-launch = idle: kernel-launch (CPU dispatch)
- I-sync = idle: sync wait
- I-copy = idle: H↔D copy
- OTH = others
- OVL = overlapped